

Year 7 • Chapter 7 • Decimals: answers (teacher copy)

7.01 Ordering decimals (booklet p. 3)

WE DO examples

- Ex 1: 0.7
- Ex 2: 34.89
- Ex 3: 8 hundredths
- Ex 4: 3.998, 34.89, 34.98, 36.86, 38.141, 340
- Ex 5: 1.293, 1.239, 1.1015, 1.015, 1.006
- Ex 6: e.g. 2.355

Set A

- 0.7
- 2.5
- 0.1
- 14.3
- 5.6
- 0.34

Set B

- 7 tenths
- 7 hundredths
- 7 thousandths
- 7 tens
- 7 thousandths
- 7 ones

Set C

- 3.998, 34.89, 34.98, 36.86, 38.141, 340
- 0.045, 0.405, 0.45, 0.5, 0.54
- 2.01, 2.011, 2.1, 2.101, 2.11
- 7.06, 7.066, 7.6, 7.606, 7.66

Set D

- 1.293, 1.239, 1.1015, 1.015, 1.006
- e.g. 2.355
- 12.018 s
- 0.1: 0.100 000 > 0.099 999

Extension

- 0.605
- 9 (4.31 to 4.39)

Exit ticket

- Level 1: 0.4
- Level 2: 5.02, 5.2, 5.22
- Level 3: 0.33, 0.303, 0.3, 0.033

7.02 Decimals and fractions (booklet p. 7)

WE DO examples

- Ex 1: $\frac{1}{2}$
- Ex 2: $\frac{89}{100}$
- Ex 3: $\frac{111}{250}$
- Ex 4: 0.4
- Ex 5: 0.07
- Ex 6: 0.011

Set A

- $\frac{3}{10}$
- $\frac{89}{100}$
- $\frac{9}{100}$
- $\frac{7}{10}$
- $\frac{21}{100}$
- $\frac{3}{1000}$

Set B

- 0.4
- 0.13
- 0.07
- 0.011
- 0.9
- 0.125

Set C

- $\frac{1}{2}$
- $\frac{1}{4}$
- $\frac{111}{250}$
- $\frac{2}{25}$
- $2\frac{3}{5}$
- $3\frac{1}{8}$

Set D

- 0.6
- 0.35
- 0.036
- $\frac{3}{8} = 0.375$ is larger

Extension

- 0.075
- $4\frac{3}{40}$

Exit ticket

- Level 1: $\frac{7}{10}$
- Level 2: 0.23
- Level 3: $\frac{7}{20}$

7.03 Adding and subtracting decimals

(booklet p. 11)

WE DO examples

- Ex 1: 16.88
- Ex 2: 2.379
- Ex 3: 27.595
- Ex 4: 104.49
- Ex 5: 396.51
- Ex 6: 65.355

Set A

1. 5.9
2. 16.88
3. 1.5
4. 4.5
5. 7.3
6. 5.75

Set B

1. 2.379
2. 5.18
3. 4.55
4. 12.03
5. 23.568
6. 19.98

Set C

1. 104.49
2. 1 037.9
3. 11.315
4. 177.595
5. 252.62
6. 5.544

Set D

1. 65.355
2. 396.51
3. \$29.75
4. 0.85 kg

Extension

1. 8.43
2. $6.42 + 5.31 = 11.73$ (or similar)

Exit ticket

1. Level 1: 4.3
2. Level 2: 6.55
3. Level 3: 19.08

7.04 Multiplying and dividing by powers of 10 (booklet p. 15)

WE DO examples

1. Ex 1: 75.4
2. Ex 2: 754
3. Ex 3: 7 540
4. Ex 4: 1.39
5. Ex 5: 0.139
6. Ex 6: 0.0139

Set A

1. 75.4
2. 754
3. 3
4. 4 820
5. 5.6
6. 125

Set B

1. 1.39
2. 0.139
3. 0.6
4. 0.0452
5. 0.008
6. 0.35

Set C

1. 0.0139
2. 7.2
3. 25 000
4. 0.00091
5. 4
6. 850

Set D

1. \$135
2. \$8.75
3. 0.096 mm
4. 1000

Extension

1. 7 200
2. 0.16

Exit ticket

1. Level 1: 38
2. Level 2: 0.526
3. Level 3: 49

7.05 Using a known product (booklet p. 19)

WE DO examples

1. Ex 1: 319.2
2. Ex 2: 31.92
3. Ex 3: 319.2
4. Ex 4: 3.192
5. Ex 5: 319.2
6. Ex 6: 0.3192

Set A

1. 319.2
2. 319.2
3. 31.92
4. 31 920
5. 31.92
6. 31.92

Set B

1. 34.5
2. 3.45
3. 3.45
4. 3.45
5. 345
6. 0.0345

Set C

1. 319.2
2. 3.192
3. 0.3192
4. 3.192
5. 31.92
6. 42

Set D

1. 45
2. 45
3. 0.045
4. 12.5

Extension

1. 17
2. 24

Exit ticket

1. Level 1: 32.2
2. Level 2: 3.22
3. Level 3: 0.0322

7.06 Multiplying decimals (booklet p. 23)

WE DO examples

1. Ex 1: 16.5
2. Ex 2: 0.6
3. Ex 3: 7.32
4. Ex 4: 1.3325
5. Ex 5: 0.001
6. Ex 6: 5.137

Set A

1. 16.5
2. 4.6
3. 29.1
4. 288.92
5. 1.8
6. 12

Set B

1. 0.6
2. 0.12
3. 0.63
4. 0.5
5. 0.001
6. 1.21

Set C

1. 7.32
2. 1.3325
3. 5.137
4. 4.34
5. 2.1
6. 0.543

Set D

1. 14.4 m²
2. \$80.33
3. 0.008
4. less: multiplying by less than 1 makes it smaller

Extension

1. 0.0081
2. e.g. 0.4×0.6

Exit ticket

1. Level 1: 8.6
2. Level 2: 0.42
3. Level 3: 3.29

7.07 Dividing decimals by whole numbers

(booklet p. 27)

WE DO examples

1. Ex 1: 381.2
2. Ex 2: 12.2
3. Ex 3: 65.2
4. Ex 4: 0.6875
5. Ex 5: 0.4375
6. Ex 6: \$8.40

Set A

1. 381.2
2. 2.1
3. 3.1
4. 0.25
5. 0.09
6. 2.1

Set B

1. 12.2
2. 65.2
3. 0.91
4. 0.08
5. 7.8
6. 0.9

Set C

1. 0.6875
2. 0.4375
3. 1.75
4. 0.15
5. 2.42
6. 0.15

Set D

1. \$8.40
2. 1.125 m
3. \$1.45
4. 4.22 km

Extension

1. 5.6
2. 1.25 and 0.625: doubling the divisor halves the answer

Exit ticket

1. Level 1: 3.4
2. Level 2: 2.4
3. Level 3: 1.325

7.08 Dividing by a decimal (booklet p. 31)

WE DO examples

1. Ex 1: 25
2. Ex 2: 31.8
3. Ex 3: 230
4. Ex 4: 25.4
5. Ex 5: 5
6. Ex 6: 14 cups

Set A

1. 25
2. 30
3. 9
4. 8
5. 6
6. 20

Set B

1. 31.8
2. 4.9
3. 32
4. 0.8
5. 30
6. 10

Set C

1. 230
2. 25.4
3. 5
4. 49
5. 600
6. 0.42

Set D

1. 14 cups
2. 12.5 km/L
3. 14 m
4. there are 10 halves in 5

Extension

1. 180
2. the same: both are 80

Exit ticket

1. Level 1: 20
2. Level 2: 9
3. Level 3: 4

7.09 Fractions to decimals (booklet p. 35)

WE DO examples

1. Ex 1: 0.8
2. Ex 2: 0.375
3. Ex 3: $0.\dot{6}$
4. Ex 4: $0.\dot{5}$
5. Ex 5: $0.4\dot{5}$
6. Ex 6: $0.1\dot{6}$

Set A

1. 0.8
2. 0.25
3. 0.75
4. 0.5
5. 0.4
6. 0.35

Set B

1. 0.375
2. 0.625
3. 0.0625
4. 0.225
5. 0.875
6. 0.12

Set C

1. $0.\dot{6}$
2. $0.\dot{5}$
3. $0.\dot{3}$
4. $0.4\dot{5}$
5. $0.1\dot{6}$
6. $0.57142\dot{8}$

Set D

1. 0.6, $\frac{5}{8}$, $\frac{2}{3}$, 0.67
2. $0.\dot{1}$, $0.\dot{2}$, $0.\dot{3}$; $0.\dot{7}$
3. 0.142857 repeating: 6 digits
4. halves, quarters, fifths, eighths, tenths

Extension

1. $\frac{7}{9}$
2. $\frac{2}{11}$

Exit ticket

1. Level 1: 0.6
2. Level 2: 0.875
3. Level 3: $0.\dot{3}$

7.10 Rounding decimals (booklet p. 39)

WE DO examples

1. Ex 1: 406.3
2. Ex 2: 125.72
3. Ex 3: 3.90
4. Ex 4: 2345.9
5. Ex 5: 78654.06
6. Ex 6: 678.431

Set A

1. 406.3
2. 3.1
3. 12.4
4. 1.0
5. 7.1
6. 2345.9

Set B

1. 125.72
2. 3.90
3. 0.56
4. 20.00
5. 4.10
6. 78654.06

Set C

1. 678.431
2. 6
3. 0.05
4. 13.00
5. 0.3
6. 10.0

Set D

1. 0.67
2. \$15.74
3. 4.65 and 4.74
4. 3.1, 3.14, 3.142

Extension

1. 23 000
2. 0.0061

Exit ticket

1. Level 1: 5.7
2. Level 2: 12.35
3. Level 3: 0.900

7.11 Decimals and money (booklet p. 43)

WE DO examples

1. Ex 1: \$470.20
2. Ex 2: \$1.34
3. Ex 3: \$21.75
4. Ex 4: \$38.20
5. Ex 5: \$17.85
6. Ex 6: \$8.66

Set A

1. \$8.25
2. \$13.15
3. \$12.00
4. \$6.20
5. \$62.55
6. \$7.80

Set B

1. \$17.85
2. \$4.10
3. \$10.00
4. \$25.05
5. \$0.50
6. \$13.35

Set C

1. \$470.20
2. \$1.34
3. \$8.66
4. \$12.00
5. \$42.45
6. \$52.00

Set D

1. \$38.20
2. \$44.60
3. 2 L for \$4.60 (\$2.30/L vs \$2.40/L)
4. \$48.40

Extension

1. one of each pack: 8 pens for \$10.70
2. Yes: \$1.20 per pen

Exit ticket

1. Level 1: \$14.60
2. Level 2: \$5.65
3. Level 3: \$12.75